A gene that makes its possessors die is called a lethal gene. A semilethal gene has some debilitating effect, such that it makes death from other causes more probable. Any gene exerts its maximum effect on bodies at some particular stage of life, and lethals and semilethals are not exceptions. Most genes exert their influence during foetal life, others during childhood, other during young adulthood, others in middle age, and yet others in old age. (Reflect that a caterpillar and the butterfly it turns into have exactly the same set of genes.) Obviously lethal genes will tend to be removed from the gene pool. But equally obviously a late-acting lethal will be more stable in the gene pool than an early-acting lethal. A gene that is lethal in an older body may still be successful in the gene pool, provided its lethal effect does not show itself until after the body has had time to do at least some reproducing. For instance, a gene that made old bodies develop cancer could be passed on to numerous offspring because the individuals would reproduce before they got cancer. On the other hand, a gene that made young adult bodies develop cancer would not be passed on to very many offspring, and a gene that made young children develop fatal cancer would not be passed on to any offspring at all. According to this theory then, senile decay is simply a by-product of the accumulation in the gene pool of late-acting lethal and semi-lethal genes, which have been allowed to slip through the net of natural selection simply because they are late-acting.

The aspect that Medawar himself emphasizes is that selection will favour genes that have the effect of postponing the operation of other, lethal genes, and it will also favour genes that have the effect of hastening the effect of good genes. It may be that a great deal of evolution consists of genetically-controlled changes in the time of onset of gene activity.

It is important to notice that this theory does not need to make any prior assumptions about reproduction occurring only at certain ages. Taking as a starting assumption that all individuals were equally likely to have a child at any age, the Medawar theory would quickly predict the accumulation in the gene pool of late-acting deleterious genes, and the tendency to reproduce less in old age would follow as a secondary consequence.

As an aside, one of the good features of this theory is that it leads us to some rather interesting speculations. For instance it follows from it that if we wanted to increase the human life span, there are two general